Second Terminal Examination - 2079

Class : XI **Subject: Physics (1011)** **Full Marks: 75**

Time: 3 hrs. **SET 'A'** **Pass Marks: 30**

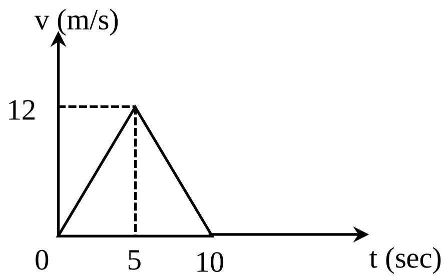
Candidates are required to give their answers in their own words as far as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

Group A

Rewrite the best alternative of each question in your answer sheet. [11 × 1 = 11]

1. Which is a vector quantity?
   a) Electric charge b) Electrostatic potential
   c) Electric current d) Electric field intensity
2. If force $\vec{F} = \hat{\imath} + \hat{\jmath} + \hat{k}$ and displacement $\vec{S} = 3\hat{\imath} + 2\hat{\jmath} + 5\hat{k}$, then work done by force is
   a) 0 b) 10 unit c) 50 unit d) 8 unit
3. A body is projected upward with a velocity of 100m/s. It will strike the ground after
   a) 10s b) 20s c) 15s d) 5s
4. The velocity time graph of a runner is shown in the figure,



   The distance travelled by the runner between t = 0s to t = 10s is
   a) 60m b) 30m c) 20m d) 50m

5. The number of significant figures in 7316000 are

a) 7 b) 6 c) 5 d) 4

6. A body of mass 400 g and specific heat 0.1 in cgs system of units has a water equivalent

a) 400 g b) 40 g c) 4 g d) 0.4 g

7. An object is placed at a distance $\frac{f}{2}$ from the pole of a convex mirror of focal length f. The image formed is

a) at $\frac{f}{2}$ virtual and erect

b) at f virtual inverted

c) at $\frac{f}{4}$ virtual and erect

d) at $\frac{f}{3}$, virtual, erect.

8. The angle of emergence at grazing emergence in prism is :

a) 0° b) 45° c) 60° d) 90°

9. The relation for deviation produced by small angled prism is :

a) $\delta = \mu - A$

b) $\delta = \mu + A$

c) $\delta = (\mu + 1) A$

d) $\delta = (\mu - 1) A$

10. A diver in water at a depth 1 m sees the whole outside world through horizontal circle of radius ..... when the refractive index is μ.

a) $\frac{1}{\mu - 1}$

b) $\frac{1}{\sqrt{\mu^2 - 1}}$

c) $\frac{\mu}{\sqrt{\mu^2 - 1}}$

d) $\frac{1}{\mu}$

11. Coulomb force between two charges kept at certain distance if F. If whole space between charges is filled with medium of dielectric constant $\varepsilon_r = 4$, then the new Coulomb's force is

a) F b) $\frac{3F}{8}$ c) $\frac{F}{4}$ d) $\frac{2F}{9}$

Group – B

Give short answer to the following questions. **[8 × 5 = 40]**

12. a. Can a quantity have unit but no dimensions? Explain. [1]

    b. A student writes an expression of force causing a body of mass 'm' to move in a circular motion with a velocity 'v' as $F = mv^2$. Check its correctness using dimensional analysis. [2]

    c. Draw the velocity-time graph for a body moving in a straight line with uniform acceleration and hence derive the relation $v = u + at$, where symbols carry their usual meanings. [2]

13 a. Show that the path of projectile projected horizontally from the top of tower is parabolic and deduce the expression for its time of flight and horizontal range. [3]

   b. A man walks towards east with velocity 3km/hr encounters rain falling vertically downwards with velocity $3\sqrt{3}$ km/hr. What is the angle at which man should incline his umbrella to protect from rain? [2]

14. a. A projectile is launched with an initial velocity of 30m/s at an angle 60° above horizontal. Calculate

    i. maximum height reached. [1]

    ii. time at which velocity and acceleration become perpendicular. [1]

    b. Can you give the example of the motion such that velocity is zero but acceleration is non-zero? [1]

    c. A ball is dropped gently from the top of the tower and another ball is thrown horizontally at same time. Which ball will hit the ground first? [2]

15. The first pendulum clock was invented in 1656 A.D. by Dutch scientist and inventor Christiaan Huygens, and patented the following year. Huygens was inspired by investigations of pendulums by Galileo Galilei beginning around 1602 A.D. Galileo discovered the key property that makes pendulums useful timekeepers.

a. Why is invar used in making a clock pendulum? [1]

b. Pendulum clocks generally run fast in winter and slow in summer, why? [1]

c. A second pendulum made of brass keeps correct time at 10^0 C. How many second it will lose or gain per day when the temperature of its surrounding rises to 35^0 C? [linear expansivity of brass = 18×10^{-6} /^{o}C] [3]

16. a. A scientific book describes a new temperature scale called Z, in which boiling and freezing points of water are referred as 65^oZ and -15^oZ respectively.

i. To what temperature on Fahrenheit scale would a temperature -95^oZ correspond. [2]

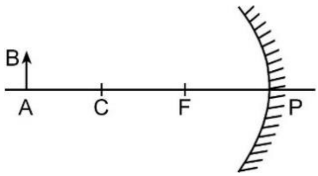
ii. What temperature change on the Z-scale would correspond to change of 40^o on Celsius scale? [1]

b. A steel wire 8m long and 4mm in diameter is fixed to two rigid supports. Calculate the increase in tension when the temperature falls by 10^0C. Given linear expansivity of steel = 12×10^{-6}/K, Young's modulus for steel = 2×10^{11} N/m^2. [2]

17. a. What is lateral shift? On what factors does the lateral shift depend? [2]

b. Derive an expression for its value. How does the lateral shift change with the increase in the angle of incidence? [3]

18. Study the diagram and answer the questions.



a. How does the nature of image vary if the object AB is moved from beyond centre towards pole of the concave mirror? Explain. [2]

b. An erect image, three times the size of the object is obtained with a concave mirror of radius of curvature 36 cm. What is the position of the object? [3]

19. a. Can a charged body attract an uncharged body? Explain. [2]

b. Riya sees a neutral conductor in a physics lab and wishes to make it negatively charged. Explain with figure, the steps to be followed by her to make the body charged negatively by method of induction. [3]

Group – C

Give long answer to the following questions. **[3 × 8 = 24]**

20. a. State the triangle law of vector addition. Using this law, find the magnitude and direction of the resultant of two vectors $\vec{P}$ and $\vec{Q}$ inclined at angle 'θ' with each other. [3]

b. Two dogs pull horizontally on the ropes attached to a post, the angle between the ropes is 60°. If dog A exerts a force of 270N and dog B exerts a force of 300N, find the magnitude of the resultant force and the angle made by resultant with dog A's rope. [3]

c. Is a physical quantity having both magnitude and direction necessarily a vector quantity? Explain. [1]

d. If a vector has zero magnitude, is it meaningful to call it a vector? [1]

21. a. Expansion of liquids is studied only in terms of volume expansion. When liquid in a container is supplied heat, first the container expands followed by expansion of liquid. So, two types of liquid expansion come into exist.

i. Define coefficient of real and apparent expansion of liquid. [2]

ii. Derive relation between them. [2]

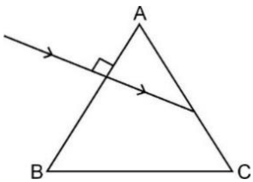
b. A copper vessel with a volume of exactly $1.80 m^3$ at a temperature 20°C is filled with glycerin. If the temperature rises to 30°C, how much glycerin will spill out? (Given: α for copper 16.7×10^{-6} °C^{-1}, γ for glycerin = 5.3×10^{-4} °C^{-1}] [3]

c. Why the glycerin is spilled out from copper vessel in above question? [1]

22 a. Why does the sun look slightly oval when it is near the horizon? [1]

b. Show that the deviation produced by small angled prism is independent of angle of incidence. [3]

c. A narrow beam of light is incident normally on one face of an equilateral prism (μ = 1.45) as shown. Show with reason how the ray emerges from the prism. [2]



d. If the same prism is surrounded by water (μ = 1.33), what is the angle of emergence? [2]

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